Machine Learning Model Based on Sentiment Analysis

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Objective

- Build a Machine Learning model
  - Naïve Bayes
  - Support Vector Machine
- Use model to classify the market performance
  - Data: wallstreetbets.csv containing Reddit comments and S&P 500 next day market returns
  - Target Variable: Market Performance
  - Performance measure: Accuracy

Methodology 1: Naïve Bayes - Sentiment Extraction

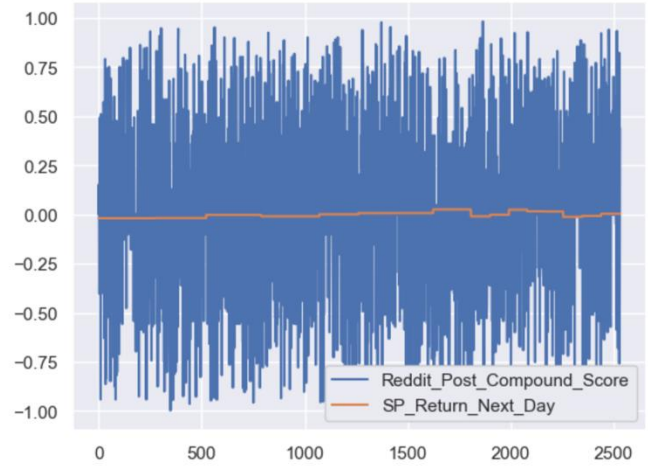
- The sentiment analysis assigns sentiment scores to each Reddit post based on its content.
- The neg, neu, and pos scores represent the proportion of negative, neutral, and positive sentiments in the post, respectively.
- The compound score combines all sentiment information into a single value, representing the overall sentiment of the text.

	Reddit_Post	neg	neu	pos	compound
0	tomorrow	0.00	1.00	0.00	0.00
1	yeah much pussy hold option always expire wort...	0.22	0.53	0.25	0.15
2	buy put portfolio	0.00	1.00	0.00	0.00
3	sound like time waste get shoot tbh	0.44	0.34	0.21	-0.40
4	find vin diesel paul walker way quite bx low k...	0.18	0.82	0.00	-0.34
...	...	...	...	...	...
2526	honest question gme fell since open market gen...	0.00	0.58	0.42	0.82
2527	gme could go time bulgarian boy able stop ape ...	0.28	0.54	0.18	-0.76
2528	lol sell little loss market want dump bad	0.54	0.20	0.26	-0.65
2529	soft land get price market curve inversion deepen	0.00	1.00	0.00	0.00
2530	better serve spend time learn interpret balanc...	0.00	0.79	0.21	0.44

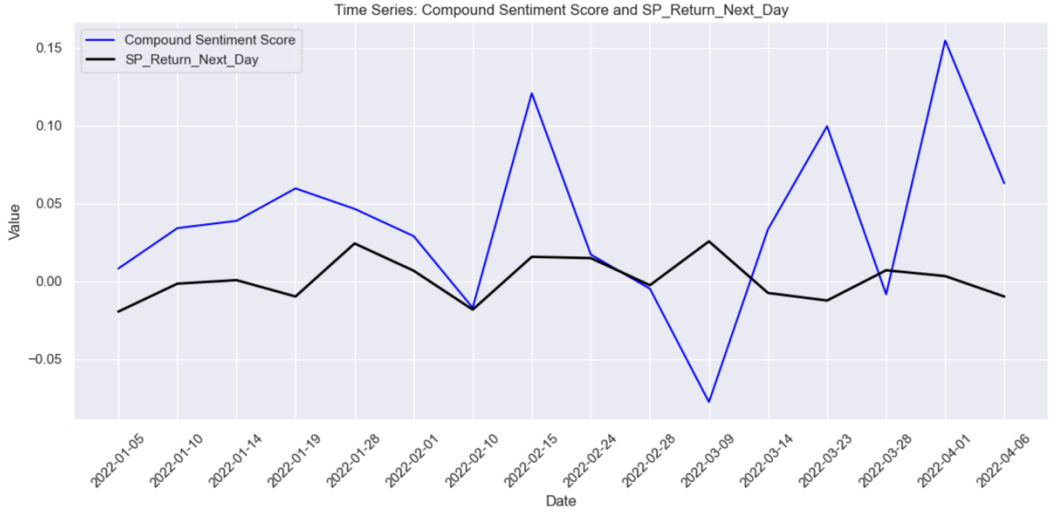
2531 rows × 5 columns

Sentiment Flow and Data Exploration

- Connect sentiment score for each reddit comment to S&P 500 returns the next day
- Calculate mean sentiment score for the day of the comment and connect to the returns next day
- A negative correlation suggests that there might be a weak inverse relationship between the sentiment score and the stock market returns for the next day.




```
neg        0.14
neu        0.69
pos        0.17
compound   0.02
```




```
Correlation between Compound Sentiment Score and SP_Return_Next_Day: -0.10571495263844403
```

Results: Naïve Bayes- Comment Level

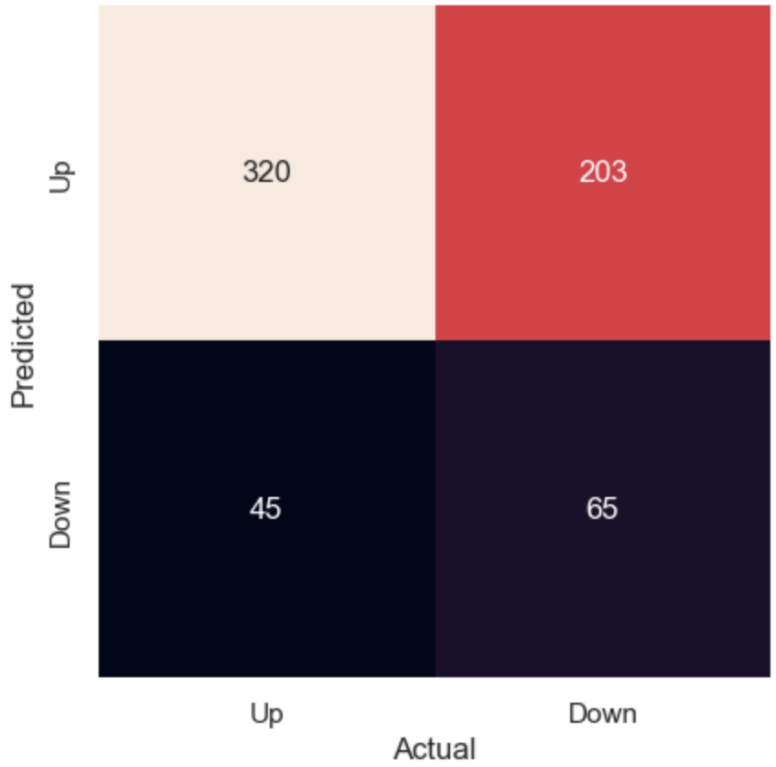
- **Accuracy**: The model's accuracy is approximately 60.82% meaning that the model's predictions are correct for about 60.82% of the instances in the dataset

- **Precision**: The model's predictions of 'Up' are correct for approximately 61.19% of the instances predicted as 'Up'. For the 'Down' class, the model's predictions of 'Down' are correct for approximately 59.09% of the instances predicted as 'Down'.

- **Recall**: For the 'Up' class, the model correctly identifies approximately 87.67% of the actual 'Up' instances. For the 'Down' class, the model correctly identifies approximately 24.25% of the actual 'Down' instances.

- The model seems to have higher recall for the 'Up' class, indicating it's better at identifying 'Up' instances, but lower recall for the 'Down' class, suggesting it may struggle with 'Down' instances.

```
Model Accuracy: 0.608214849921011
Precision: [0.61185468 0.59090909]
Recall: [0.87671233 0.24253731]
```



Methodology 2: Naïve Bayes – Day Average

- The DataFrame is grouped by the 'Date' column and then the mean of the sentiment metrics is calculated for each group.

- Each row in the DataFrame represents aggregated sentiment analysis metrics for a specific date. For example, on January 5, 2022, the sentiment analysis metrics are as follows:
  - Positive sentiment score ('pos'): 0.14
  - Neutral sentiment score ('neu'): 0.72
  - Negative sentiment score ('neg'): 0.13
  - Compound sentiment score ('compound'): 0.01

Date	pos	neu	neg	compound
2022-01-05	0.14	0.72	0.13	0.01
2022-01-10	0.17	0.66	0.17	0.03
2022-01-14	0.17	0.71	0.13	0.04
2022-01-19	0.16	0.71	0.13	0.06
2022-01-28	0.18	0.68	0.15	0.05
2022-02-01	0.17	0.67	0.16	0.03
2022-02-10	0.15	0.67	0.17	-0.02
2022-02-15	0.23	0.68	0.08	0.12
2022-02-24	0.21	0.63	0.16	0.02
2022-02-28	0.15	0.72	0.12	-0.00
2022-03-09	0.14	0.69	0.17	-0.08
2022-03-14	0.18	0.67	0.15	0.03
2022-03-23	0.18	0.72	0.10	0.10
2022-03-28	0.17	0.66	0.17	-0.01
2022-04-01	0.24	0.69	0.07	0.15
2022-04-06	0.21	0.66	0.13	0.06

Results: Naïve Bayes - Day Average

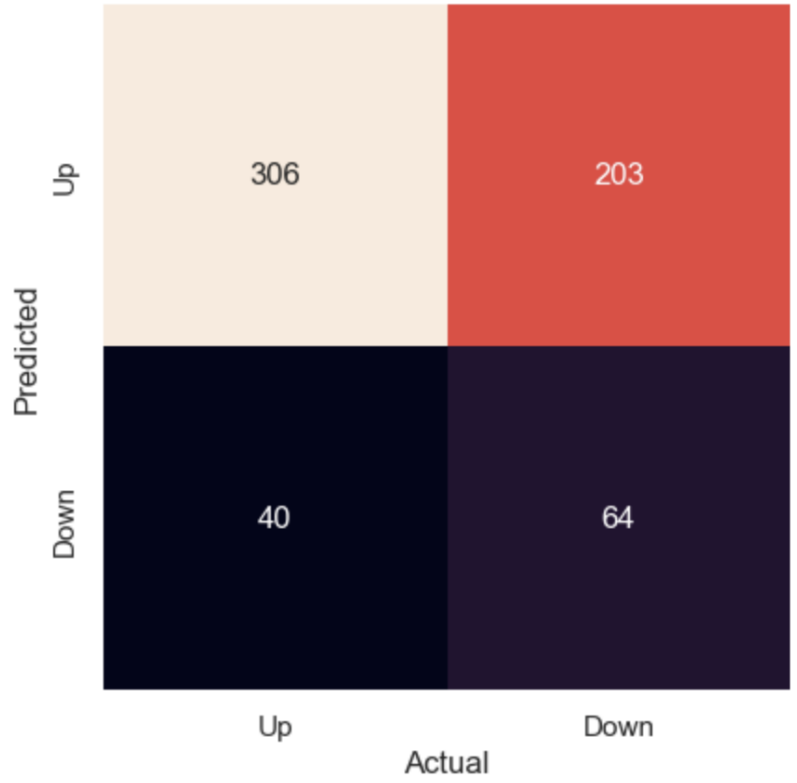
- **Accuracy**: The model's accuracy is approximately 60.36% meaning that the model's predictions are correct for about 60.36% of the instances in the dataset

- **Precision**: The model's predictions of 'Up' are correct for approximately 60.18% of the instances predicted as 'Up'. For the 'Down' class, the model's predictions of 'Down' are correct for approximately 61.54% of the instances predicted as 'Down'.

- **Recall**: For the 'Up' class, the model correctly identifies approximately 88.44% of the actual 'Up' instances. For the 'Down' class, the model correctly identifies approximately 23.97% of the actual 'Down' instances.

- The model seems to have higher recall for the 'Up' class, indicating it's better at identifying 'Up' instances, but lower recall for the 'Down' class, suggesting it may struggle with 'Down' instances.

```
Model Accuracy: 0.6035889070146819
Precision: [0.60117878 0.61538462]
Recall: [0.88439306 0.23970037]
```



Methodology 3: Support Vector Machine - Sentiment Extraction

- Transforms the text data into a numerical format suitable for machine learning models
- Each row represents a Reddit post, and the columns represent individual words or terms found in the posts.
- The values in the DataFrame indicate whether a specific word is present (True) or absent (False) in each post.

	back	bot	buy	call	come	day	even	fuck	get	go	...	stock	take	think	time	today	would	year	SP_Return_Next_Day	Market	Date
1	False	False	False	False	False	False	False	False	False	False	...	False	False	False	True	False	False	False	-0.02	Down	2022-01-05
2	False	False	True	False	False	False	False	False	False	False	...	False	False	False	False	False	False	False	-0.02	Down	2022-01-05
3	False	False	False	False	False	False	False	False	True	False	...	False	False	False	True	False	False	False	-0.02	Down	2022-01-05
5	False	False	False	False	False	False	False	False	False	False	...	False	False	False	False	False	False	False	-0.02	Down	2022-01-05
9	False	False	False	False	False	False	False	True	False	True	...	False	False	False	False	False	False	False	-0.02	Down	2022-01-05
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
2526	False	False	False	False	False	True	False	False	False	False	...	False	False	False	False	False	False	False	0.00	Up	2022-04-01
2527	False	False	False	False	False	True	False	True	False	True	...	False	False	False	True	False	False	False	0.00	Up	2022-04-01
2528	False	False	False	False	False	False	False	False	False	False	...	False	False	False	False	False	False	False	0.00	Up	2022-04-01
2529	False	False	False	False	False	False	False	False	True	False	...	False	False	False	False	False	False	False	0.00	Up	2022-04-01
2530	False	False	False	False	False	False	False	False	False	False	...	False	False	False	True	False	False	False	0.00	Up	2022-04-01

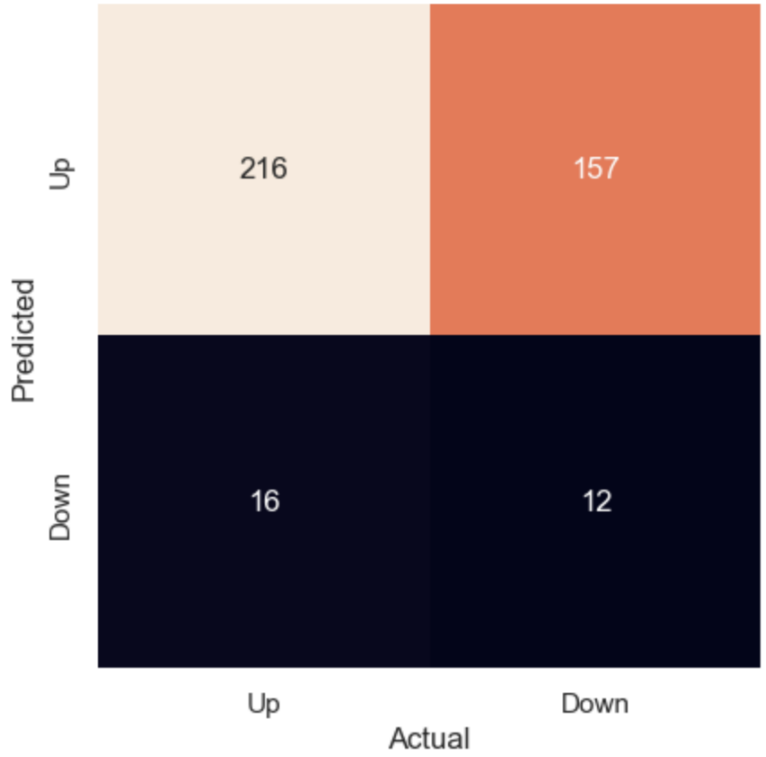
Results: Support Vector Machine

- **Accuracy**: The model's accuracy is approximately 56.86% meaning that the model's predictions are correct for about 56.86% of the instances in the dataset

- **Precision**: The model's predictions of 'Up' are correct for approximately 57.91% of the instances predicted as 'Up'. For the 'Down' class, the model's predictions of 'Down' are correct for approximately 42.86% of the instances predicted as 'Down'.

- **Recall**: For the 'Up' class, the model correctly identifies approximately 93.1% of the actual 'Up' instances. For the 'Down' class, the model correctly identifies approximately 71.06% of the actual 'Down' instances.

- The model seems to have relatively high recall for the 'Up' class but relatively low precision and recall for the 'Down' class. This suggests that the model may be better at identifying 'Up' instances but struggles with 'Down' instances.




```
Model Accuracy: 0.5685785536159601
Precision: [0.57908847 0.42857143]
Recall: [0.93103448 0.07100592]
```

Conclusion

- For the 'Accuracy' metric, 'Naïve Bayes: Individual Comment' and 'Naïve Bayes: Mean' outperform the SVM model.
- For the 'Precision' metric, 'Naïve Bayes: Mean' has the highest precision for both 'Up' and 'Down' classes, while 'SVM' has the lowest precision for the 'Down' class.
- For the 'Recall' metric, 'SVM' has the highest recall for the 'Up' class, and 'Naïve Bayes: Individual Comment' has the highest recall for the 'Down' class.
- Overall, 'Naïve Bayes: Individual Comment' and 'Naïve Bayes: Mean' models generally perform better than the 'SVM' model in terms of accuracy and precision, with 'SVM' having a notably high recall for the 'Up' class but lower recall for the 'Down' class.

Model	Market	Accuracy	Precision	Recall
Naïve Bayes: Individual Comment	Up	0.6082	0.6119	0.8767
	Down		0.5909	0.2425
Naïve Bayes: Mean	Up	0.6036	0.6012	0.8844
	Down		0.6154	0.2397
SVM	Up	0.5686	0.5791	0.9310
	Down		0.4286	0.0710

Appendix

- Please find code attached to the submission

Thank You